Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/26/2015
PI Name:	Roux, Stanley Ph.D.		
Project Title:	Rapid Signaling Changes Induced by Gravity in Cells of the Fern Ceratopteris richardii		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 per NSSC information (Ed., 5/26/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Greg Ph.D. (University of Texas)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Single-celled spores of the fern Ceratopteris richardii respond within seconds to both the direction and magnitude of gravity stimuli by altering the direction and magnitude of a trans-cell Ca2+ current that is required for the gravity response. This proposal aims to clarify the molecular mechanisms by which gravity alters the trans-cell current, and identify key signaling steps that link the current to cellular polarity changes directed by gravity. This focus on gravity signaling addresses an overarching question in the Space Biology Science Plan. Ca2+-uptake channels and Ca2+-ATPase efflux pumps drive the Ca2+ current across spore cells. Transcriptomic and proteomic analyses of genes and proteins highly expressed in spores at the peak of the Ca2+ current have identified likely candidates for the pumps and channels. Immunological approaches and yeast-based activity assays will be used to determine whether these transporters are localized on the plasma membrane and are regulated post-translationally. Recent evidence indicates that extracellular eATP [eATP], a newly discovered signaling agent, helps control the gravity response in Ceratopteris, just as it helps regulate gravitropism in Arabidopsis roots. Because eATP controls the activity of Ca2+ channels in plants, we will test the regulatory link between eATP and the activity of Ca2+ channels that is crucial for the gravity response in Ceratopteris. Ca2+-binding proteins such as calmodulins, annexins, and Ca2+-dependent protein kinases are highly expressed in Ceratopteris at the peak of the Ca2+ current. They are the most likely signal transducers that help mediate the downstream cell polarization events guided by gravity, and their regulatory role in the gravity response will be tested. Expected results will generate hypotheses that predict molecular responses likely to occur in Ceratopteris cells in response to the micro-g environment, and the proposal will describe International Space Station (ISS) experiments that could test these predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The germinating spore of Ceratopteris is an especially valuable model system for studying the cellular bases of gravity-induced graviresponses on Earth, because both the sensing of gravity and the response occur in the same cell. On the cellular level mechanisms of sensory responses are very similar in plant and animal cells, thus the bottom-to-top Ca2+ current that develops early during spore germination as one of the first polarized events induced by gravity in the cell may be an early event also in animal cells known to respond to gravity, such as muscle and bone cells. Our investigation into candidate molecules likely to be associated with both the efflux and influx phases of this current will help clarify how this current is controlled. A number of studies have documented the importance of Ca2+ currents for polarized growth in both animal and plant cells; however, there are few prior studies that have identified and analyzed the molecular components of these currents. Moreover, while it is generally accepted that Ca2+ is a second messenger that participates in a myriad of plant and animal sensory responses, there is still little known about the molecular identity of the Ca2+ channels that mediate many of these responses, especially responses to gravitational stimuli. The proposed tests of Ca2+ pump and channel function and regulation, including possibly channel regulation by eATP, will not only provide novel insights into how the gravity-directed Ca2+ current originates, but also further advance an understanding of channel structures. The results will set the stage for future assays of post-translational changes that occur during space flight in the Ca2+-ATPase pumps and/or mechanosensitive channels in fern and possibly also in some animal cells when the transporter activities go from near zero in micro-gravity to quite high as the g-force rises to 2-g or higher levels achieved during parabolic flight or on a centrifuge on the ISS.
Task Progress:	Year two update on progress of specific aims Aim 1. Use proteomic, immunological, and yeast-based activity assays to determine whether cation transporters that are known to be expressed at the peak of the trans-cell Ca2+ current are localized on the plasma membrane and are regulated post-translationally. Progress: Prior work has identified a specific calcium pump, CrACA1, as playing a major role in driving the trans-cell calcium current that is directed by gravity in fern spores cells (Bushart et al., 2013; Bushart et al., 2014). We have hypothesized that a second critical transporter for the trans-cell calcium current would be a plasma membrane localized mechanosensitive cation transporter. As a prerequisite to developing molecular tools to test this hypothesis, we have obtained the full-length sequence of a putative MscS (mechanosensitive channel, small current) gene, and we are now investigating whether it actually has Msc activity by testing whether it can rescue a bacterial knockout of Msc. We also have partial sequences of two other Msc candidates in Ceratopteris and are trying to obtain full-length sequences of them by sequencing BAC-clone DNA. If any of these candidates prove to be mechanosensitive channels we will test whether the protein it encodes is modified post-translationally by gravity stimulation. Dr. Mari Salmi, the research scientist working on this project, completed the 2014 workshop on Plant Quantitative Proteomics at the University of Wisconsin in July 2014. This workshop gave her training in mass spectrometric techniques, post-translational modification identification, and global proteome quantification. The knowledge and skills gained at this workshop will be applied to evaluation of the proteins involved in regulating the trans-cellular calcium current. Aim 2. Test the role of Ca2+-binding proteins expressed at the peak of the trans-cell Ca2+ current by suppressing their expression using transformation methods. Progress: We have successfully generated transgenic Ceratopteris expressing the Calcium reporter Yellow Cameleon, and, using the same transformation methods, we are mid-way toward obtaining other transgenic Ceratopteris sporophytes that are constitutively expressing either a CaATPase or a mechanosensitive channel protein. We believe the channel and the Ca2+ATPase pump we characterized last year are likely to play a major role in generating the trans-cell calcium current that is directed by gravity. Of course we will be using mutant spores modified in their expression of these transporters to clarify how calcium transport mediates the gravity response in germinating spore cells. Aim 3. Test the role of eATP in promoting the critical Ca2+-uptake step of the spore gravity response, using extracellular luciferase reporters to assay whether the gravity stimulus induces a bottom-to-top gradient of eATP, and by intracellular reporters to assay changes in [Ca2+]cyt. Progress: In collaboration with Eric McLamore at the University of Florida, we have used ATP-selective microelectrodes to confirm that gravity induces an asymmetric gradient of eATP, with 6X higher [eATP] along the bottom of the spore. Further, we have shown that if this gradient is disrupted by eATP hydrolysis or by flooding the media with extracellular nucleotides, gravity-directed cell polarization is disrupted. Ashley Cannon (grad student) reported on this work in a mini-symposium talk at the American Society of Plant Biologists (ASPB) meeting in Minneapolis this summer. The manuscript describing the construction of the ATP-selective microelectrode and our use

	of it to measure the [eATP] of roots and spores is now published. Bushart TJ, Cannon A, Clark G, Roux SJ (2014) Structure and function of CrACA1, the major PM-type Ca2+-ATPase, expressed at the peak of the gravity-directed trans-cell calcium current in spores of the fern Ceratopteris richardii. Plant Biology 16: 151-157. Bushart TJ, Cannon AE, ul Haque A, San Miguel P, Mostajeran K, Clark GB, Porterfield DM, Roux SJ (2013) RNA-seq analysis identifies potential modulators of gravity response in spores of Ceratopteris (Parkeriaceae): Evidence for modulation by calcium pumps and apyrase activity. American Journal of Botany 100: 161-174.
Bibliography Type:	Description: (Last Updated: 10/16/2023)
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Articles in Peer-reviewed Journals	Cannon AE, Salmi ML, Bushart TJ, Roux SJ. "Studying molecular changes during gravity perception and response in a single cell." Methods in Molecular Biology. 2015;1309:199-207. http://dx.doi.org/10.1007/978-1-4939-2697-8_15 ; PubMed PMID: 25981777 , May-2015
Articles in Peer-reviewed Journals	Clark GB, Morgan RO, Fernandez MP, Salmi ML, Roux SJ. "Breakthroughs spotlighting roles for extracellular nucleotides and apyrases in stress responses and growth and development." Plant Science. 2014 Aug;225:107-16. Review. Epub 2014 Jun 11. http://dx.doi.org/10.1016/j.plantsci.2014.06.002 ; PubMed PMID: 25017166 , Aug-2014
Articles in Peer-reviewed Journals	Yang X, Wang B, Farris B, Clark G, Roux SJ. "Modulation of root skewing in Arabidopsis by apyrases and extracellular ATP." Plant Cell Physiol. 2015 Sep 27. [Epub ahead of print] http://dx.doi.org/10.1093/pcp/pcv134 ; PubMed PMID: 26412783 , Sep-2015